AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 8 · STUDENT LESSON

Loss functions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Pair the records. Penalize each error. Average every case.

Calculate and interpret losses while preserving sign,units,denominator and evidence status.

Keep one habit: Complete paired loss table → cancellation → pictured metric-choice trade-off → alignment/guard probes → independent new comparison.

MAIN LESSON · SECTION 1

How much error remains when signs cancel?

> Declare the residual direction,penalty and denominator before comparing models.

Learning goals

I can calculate per-example **squared error** and **mean squared error**.

I can distinguish **residual** cancellation from a low loss.

I can separate **training** loss from independent evaluation.

Recall the previous computation

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Parameters and architecture. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Composing layers. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Dot products and weighted sums. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Does knowing a network **output** tell us whether it is accurate?

> **My starting explanation**

End goal

By the end, I can explain loss functions using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Calculate per-example **squared error** and **mean squared error**

Distinguish **residual** cancellation from a low loss

Separate **training** loss from independent evaluation

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Turn every paired prediction error into a penalty

> This week uses prediction–target and ordinary MSE with no half factor.

Declare the residual and squared error

A loss assigns a numerical penalty to a **prediction** compared with a **target**. For this regression lesson, **residual** e = **prediction** - **target**, and **squared error** is e squared.

Squaring makes both positive and negative **residual**s contribute nonnegative penalties. We use ordinary **mean squared error** without a one-half factor.

Complete paired three-case calculation

Example · Target · Prediction · Residual · Squared error

> **Example:** A
> **Target:** 1
> **Prediction:** 2
> **Residual:** 1
> **Squared error:** 1

> **Example:** B

> **Target:** 3
>
> **Prediction:** 2
>
> **Residual:** -1
>
> **Squared error:** 1

> **Example:** C
>
> **Target:** 2
>
> **Prediction:** 4
>
> **Residual:** 2
>
> **Squared error:** 4

Average all cases and retain squared units

The **mean squared error**, or MSE, is (1 + 1 + 4) / 3 = 2. It summarises these three examples, not every possible future **input**. If **target**s use cm, **residual**s use cm and MSE uses square cm.

Name supplied and missing facts

> **List the given facts and one fact that is not supplied.**

Compare complete candidate vectors on identical targets

> A zero mean signed error can hide two wrong predictions.

Complete cancellation explanation

For **target**s [1, 3] and **prediction**s [2, 2], the mean **residual** is zero because 1 and -1 cancel. MSE is 1, correctly recording that both **prediction**s miss.

A mean **residual** can describe directional tendency, but it is not a substitute for an error magnitude.

Complete P/Q/baseline/Rpaired comparison

Same target vector[1, 3, 2] · Predictions · Residuals prediction–target · Squared errors · MSE over 3

> **Same target vector[1, 3, 2]:** P
>
> **Predictions:** [2, 2, 4]
>
> **Residuals prediction–target:** [1, –1, 2]
>
> **Squared errors:** [1, 1, 4]
>
> **MSE over 3:** 2

> **Same target vector[1, 3, 2]:** Q
>
> **Predictions:** [1, 2, 2]
>
> **Residuals prediction–target:** [0, –1, 0]
>
> **Squared errors:** [0, 1, 0]
>
> **MSE over 3:** 1/3

> **Same target vector[1, 3, 2]:** Fixed comparison baseline
>
> **Predictions:** [2, 2, 2]
>
> **Residuals prediction–target:** [1, –1, 0]

> **Squared errors:** [1, 1, 0]
>
> **MSE over 3:** 2/3

> **Same target vector[1, 3, 2]:** Supported R
>
> **Predictions:** [0, 3, 3]
>
> **Residuals prediction–target:** [–1, 0, 1]
>
> **Squared errors:** [1, 0, 1]
>
> **MSE over 3:** 2/3

Pairing,baseline provenance and sign conventions

The fixed[2, 2, 2]comparison vector is supplied for loss arithmetic; its training-fit history is not given. Do not infer that 2 was fitted from these evaluation targets. Pair predictions and targets by case identity, not only by list position after an independent sort. The opposite residual sign used in some earlier lessons gives the same squared/absolute losses, but here signed residual means prediction–target.

Keep the declared sign beside each calculation.

Runnable guarded MSE and its assumptions

```
def mse(predictions, targets):
    if not targets or len(predictions) != len(targets):
        raise ValueError("Need equal nonempty sequences")
    errors = [(p - t) ** 2
              for p, t in zip(predictions, targets)]
    return sum(errors) / len(errors)

print(mse([2, 2, 4], [1, 3, 2]))  # 2.0
print(mse([2, 2], [1, 3]))        # 1.0
```

The guard prevents silently ignoring an unmatched example or dividing by zero. It still assumes paired, numeric, valid records. Wrong row alignment produces a meaningless comparison even when list lengths agree.

Connect the visual and each penalty

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Change the criterion without changing the recorded errors

> A model ranking belongs to a named objective.

Current cancellation task

- **Target**s [0, 4] have **prediction**s [2, 2]. Compare mean **residual** and MSE.

Keep the procedure fixed. Explain what changed in the inputs and result.

Predict the pictured MSE-to-MAEchange

- Change: Choose MAE instead of MSE as the sole criterion.

▼ Check after predicting

Prediction: A and B tie at MAE 1.

Reason: **Squared error** penalises B's concentrated error more, while absolute errors both sum 2.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

See why concentrated error matters more under squaring

> State the objective before saying which candidate wins.

Complete picture targets,candidates and scope

Compare candidate losses with the same **target**s and denominator.

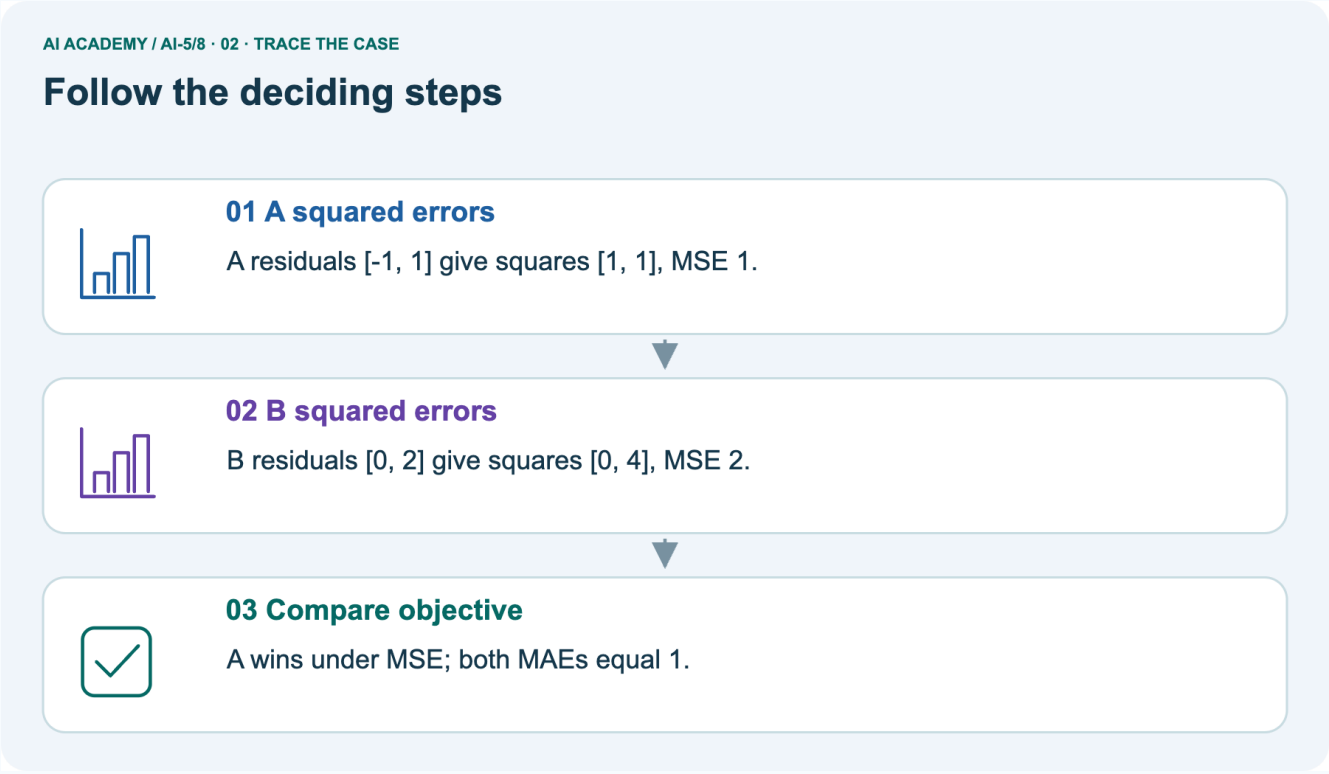
Targets [1, 3]; A predicts [0, 4]; B predicts [1, 5]. Use MSE over both cases.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate two-case fixture:targets[1, 3], A[0, 4]has residuals[–1, 1], MSE 1; B[1, 5]has[0, 2], MSE 2. Both MAEs are 1. Squaring gives the one error of size 2a penalty 4, while the two errors of size 1 give total 2. The choice A under MSE becomes a tie under MAE. Do not substitute the main three-case P/Q/Rvectors or the selected independent task's inputs.

These supplied prediction pairs are arithmetic evidence, not claimed training runs.

Trace squares,absolute values and shared denominator



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. A squared errors	A residuals [-1, 1] give squares [1, 1], MSE 1.
2. B squared errors	B residuals [0, 2] give squares [0, 4], MSE 2.
3. Compare objective	A wins under MSE; both MAEs equal 1.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

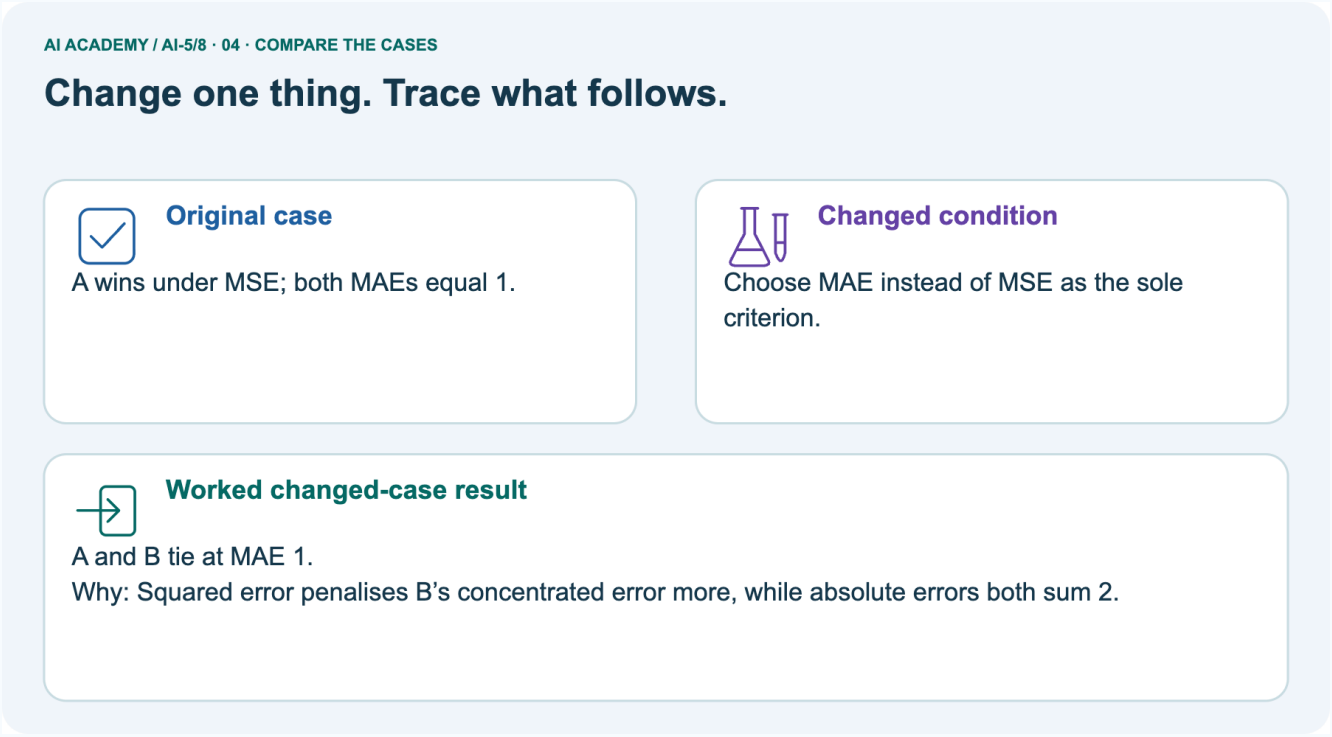
For this loss calculation, the signed error is **prediction** minus **target**. MSE squares each error before averaging; MAE takes its absolute value. A spreads its error equally across both cases, while B concentrates it in one case.

Squaring makes the larger individual error count more. State the objective before selecting a model, because these two objectives give different comparisons.

Predict the changed criterion

- Change one thing: Choose MAE instead of MSE as the sole criterion. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain metric choice and evidence limits

- **Training** loss is not automatically a held-out performance estimate or a complete measure of use consequences.

- Read the labelled picture: Why must the report name the chosen loss? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to main P/Q/baseline and try R

Use the same targets and denominator

CANDIDATE PREDICTIONS	SQUARED ERRORS	MSE
P is [2, 2, 4]	[1, 1, 4]	6 divided by 3 is 2
Q is [1, 2, 2]	[0, 1, 0]	1 divided by 3
Baseline is [2, 2, 2]	[1, 1, 0]	2 divided by 3

Each row evaluates a separate candidate against targets [1, 3, 2]. Q has the lowest MSE on these three cases. The comparison alone does not establish performance on unseen examples.

Candidate R predicts [0, 3, 3] for targets [1, 3, 2]. Calculate residuals, squared errors and MSE. Compare R with the baseline.

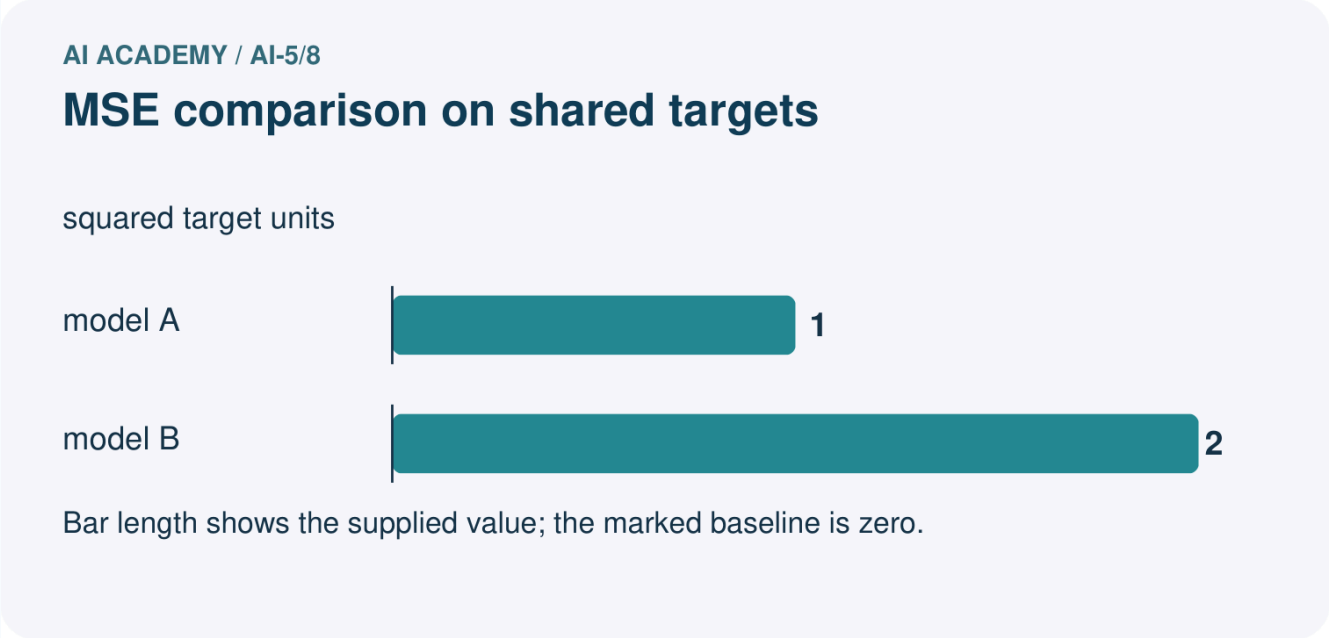
▼ **Earlier wording, alternatives and picture transcripts**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Loss functions

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/8

Loss functions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. A squared errors

A residuals [-1, 1] give squares [1, 1], MSE 1.

↓

2. B squared errors

B residuals [0, 2] give squares [0, 4], MSE 2.

↓

3. Compare objective

A wins under MSE; both MAEs equal 1.

CHANGE ONE THING

Choose MAE instead of MSE as the sole criterion.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Repair the denominator before interpreting the score

> Correct predictions still count in the number of evaluated cases.

Large errors and numerical-target assumptions

Squaring gives larger **residual**s more influence: **residual** 4 contributes 16, four times the penalty of **residual** 2. This can be useful when large misses matter, but it can also make outliers influential.

Mean absolute error uses absolute **residual**s instead; it can rank candidates differently.

MSE is appropriate here because **target**s are numerical quantities. A class identifier such as 0, 1 or 2 may merely name a category; its numeric gap does not automatically measure error severity.

Later classification models need a loss suited to their **output**s and task.

Repair the nonzero-only denominator

- A learner totals **squared error**s 1, 0, 9 and divides by 2 because there are two nonzero errors. Correct the MSE.

State one repair and the checks you would rerun.

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

Practise paired residuals and losses with less help

> Show the differences,squares,sum and divisor.

Current task and response

- For **target**s [2, 4] and **prediction**s [1, 6], record **residual**s, squares and MSE.

> **Candidate R predicts [0, 3, 3] for targets [1, 3, 2]. Calculate residuals, squared errors and MSE. Compare R with the baseline.**

> **Explain your prediction to a partner before comparing results.**

Supported W2record

Use workbook W2 for the connected example “Loss functions”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual misconception check

- Use the worked case for Loss functions. Proposed explanation: “Average **residual** is the loss”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured criterion-change reasoning

- Why must the report name the chosen loss? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the independent new comparison

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Expose cancellation

Check your reasoning

Explain the picture

Investigate cancellation,denominators and row alignment

> The complete probes distinguish arithmetic checks from valid evaluation evidence.

Current task and runnable guarded MSE

Targets [0, 4] have **prediction**s [2, 2]. Compare mean **residual** and MSE.

```
def mse(predictions, targets):
    if not targets or len(predictions) != len(targets):
        raise ValueError("Need equal nonempty sequences")
    errors = [(p - t) ** 2
              for p, t in zip(predictions, targets)]
    return sum(errors) / len(errors)

print(mse([2, 2, 4], [1, 3, 2]))  # 2.0
print(mse([2, 2], [1, 3]))        # 1.0
```

Complete normal,zero,empty,missing and pairing probes

Complete loss probe	Expected result / action
Normal predictions[2, 2, 4], targets[1, 3, 2]	MSE(1+1+4)/3=2
Cancellation predictions[2, 2], targets[0, 4]	Residuals[2, −2], mean 0, MSE 4
Perfect pair predictions[1, 3], targets[1, 3]	MSE 0on these two references
Empty predictions/targets	ValueError:Need equal nonempty sequences
Lengths2 and1	Same ValueError; never silently truncate
Missing target[None]with prediction[1]	Outside numeric/reference contract; request/review the target, not guess 0
Squares1, 0, 9 over3 cases	MSE 10/3, not 10/2
Same two predictions[2, 4], targets[1, 3]versusmispaired[3, 1]	Correct pairing MSE 1; mispaired MSE 5 even though lengths match

Complete loss probe	Expected result / action
Labels used for fitting or model choice	Report training/development loss, not an untouched final estimate

Predict,run or trace,and explain

1. Predict every probe before calling mse in separate runs or tracing it on paper. Save each residual, square, total and divisor. The function guards nonempty equal lengths but assumes valid numeric, correctly paired records; it cannot detect reordered case identities. Keep the fixed baseline's origin as supplied/unknown rather than fit it from evaluation labels. Report split, number of cases, units and MSE definition.

   A lower observed loss under one objective does not establish all future performance or real-use consequences.

2. On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Keep the evidence ledger

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or "not run"	
First mismatch and one explanation	
Boundary or missing input and expected response	

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep "not run" if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compare new predictions under two objectives

> Use the new targets and show what changes in the ranking.

Complete new independent prediction vectors

Targets are [0, 2, 4]. Model A predicts [1, 2, 3]; model B predicts [0, 2, 2]. Calculate both MSEs and identify the better model under MSE on these cases. Calculate mean absolute errors too.

State what changes in the conclusion.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Compare an unfamiliar evaluation independently

Attempt this without hints

Targets are [0, 2, 4]. Model A predicts [1, 2, 3]; model B predicts [0, 2, 2]. Calculate both MSEs and identify the better model under MSE on these cases. Calculate mean absolute errors too.

State what changes in the conclusion.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep losses linked to references and split roles

> The model's own predictions and paired targets should support the reported score.

Project artifact and evidence table

Add a labelled loss functions **evidence** sheet to Image-pattern Network Investigation. Show how you can calculate per-example **squared error** and **mean squared error**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run/trace and remaining limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with one calculated loss and one bounded claim

> Name which cases and objective the result describes.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Calculate per-example **squared error** and **mean squared error**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Target	The recorded value a model is intended to predict
Residual	Prediction minus target under the sign convention used here
Squared error	The square of a numerical residual
Mean squared error	The average squared error over the evaluated examples

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- For **target**s [2, 4] and **prediction**s [1, 6], record **residual**s, squares and MSE.
- A learner totals **squared error**s 1, 0, 9 and divides by 2 because there are two nonzero errors. Correct the MSE.
- A team tries ten models and picks the one with the lowest final-test MSE. Can that same result be presented as an untouched final evaluation?

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent three-case MSE/MAEcomparison for the first individual attempt.

Current explanation and response

- **Training** loss measures the cases used to adjust parameters. Validation loss can guide choices, so repeated use makes it development **evidence**.

  A reserved final test supports a separate evaluation claim only while it remains outside those choices. Always report the split, number of cases and loss definition.

- Next week studies local changes in a loss, preparing us to choose a direction for a parameter update.

- A team tries ten models and picks the one with the lowest final-test MSE. Can that same result be presented as an untouched final evaluation?

> **A model reports MSE zero on its training rows. What has this shown, and why is it insufficient to claim perfect future predictions?**

> **My limited claim and a useful next check**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Use the same targets and denominator

CANDIDATE PREDICTIONS	SQUARED ERRORS	MSE
P is [2, 2, 4]	[1, 1, 4]	6 divided by 3 is 2
Q is [1, 2, 2]	[0, 1, 0]	1 divided by 3
Baseline is [2, 2, 2]	[1, 1, 0]	2 divided by 3

Each row evaluates a separate candidate against targets [1, 3, 2]. Q has the lowest MSE on these three cases. The comparison alone does not establish performance on unseen examples.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Compare candidate losses with the same **target**s and denominator.

By the end, I can	How I will show it
Calculate per-example squared error and mean squared error	Explain the deciding step, show a trace or test, and state one limit.
Distinguish residual cancellation from a low loss	Explain the deciding step, show a trace or test, and state one limit.
Separate training loss from independent evaluation	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Compare candidate losses with the same **target**s and denominator.
- Which labelled step matches this detail? A wins under MSE; both MAEs equal 1.
- 1. Compare objective
- 2. A **squared error**s
- 3. B **squared error**s
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compare objective A wins under MSE; both MAEs equal 1. For this loss calculation, the signed error is **prediction** minus **target**.

MSE squares each error before averaging; MAE takes its absolute value. A spreads its error equally across both cases, while B concentrates it in one case. Squaring makes the larger individual error count more.

State the objective before selecting a model, because these two objectives give different comparisons.

Changed-case question and optional hint

- Changed-case practice: Choose MAE instead of MSE as the sole criterion. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. A and B tie at MAE 1.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: A and B tie at MAE 1. **Squared error** penalises B's concentrated error more, while absolute errors both sum 2.

Repair and retain the record

Review Parameters and architecture (ai-5/7). Goal: Distinguish architectural choices from parameter values. Short practice: How many new parameters did that one hidden unit add?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Composing layers (ai-5/5). Goal: Trace how one layer supplies the next layer **input**s. Short practice: What intermediate value explains the changed **output**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Dot products and weighted sums (ai-5/2). Goal: Calculate a dot product by matching feature positions. Short practice: Which connection explains why more right brightness lowers this score?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

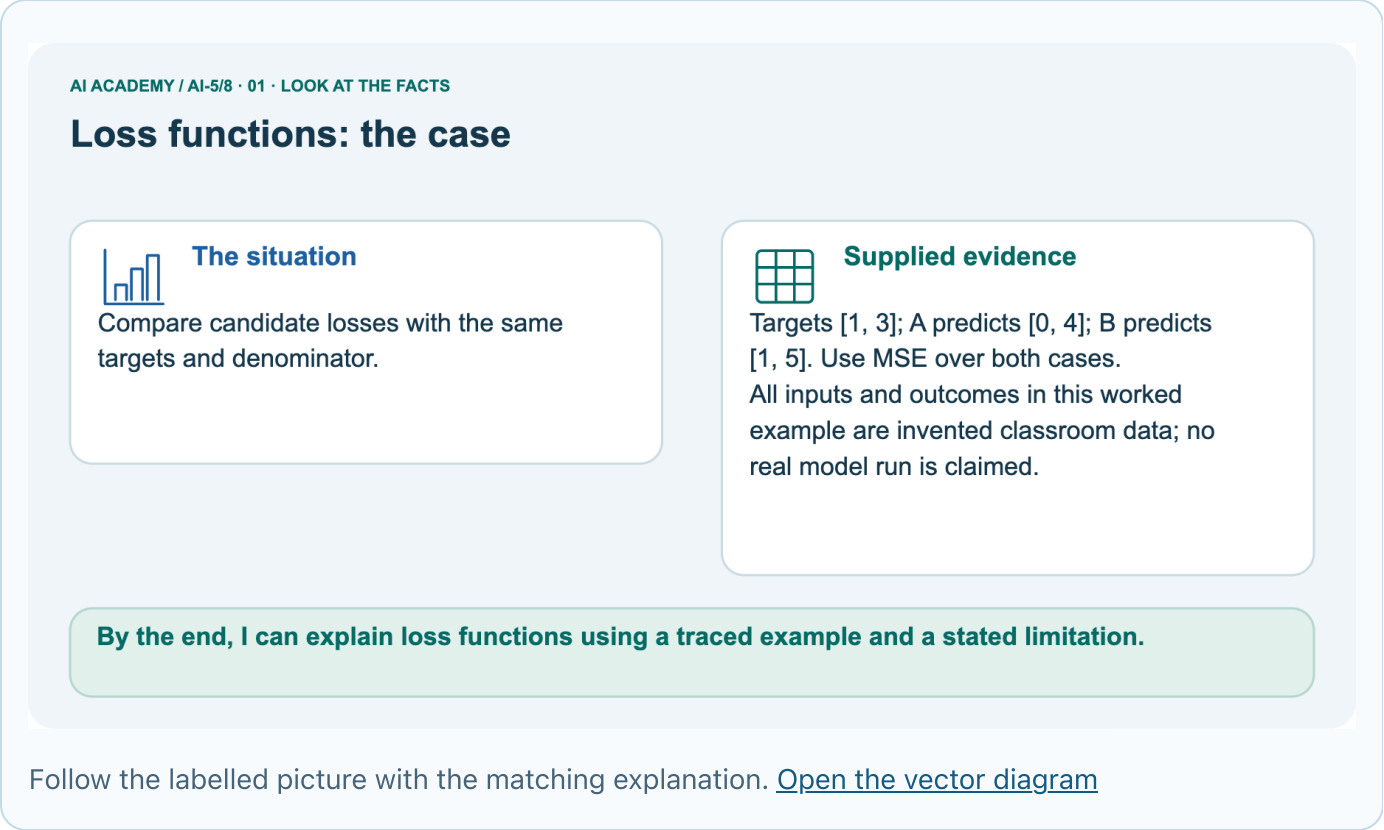
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Loss functions: the case



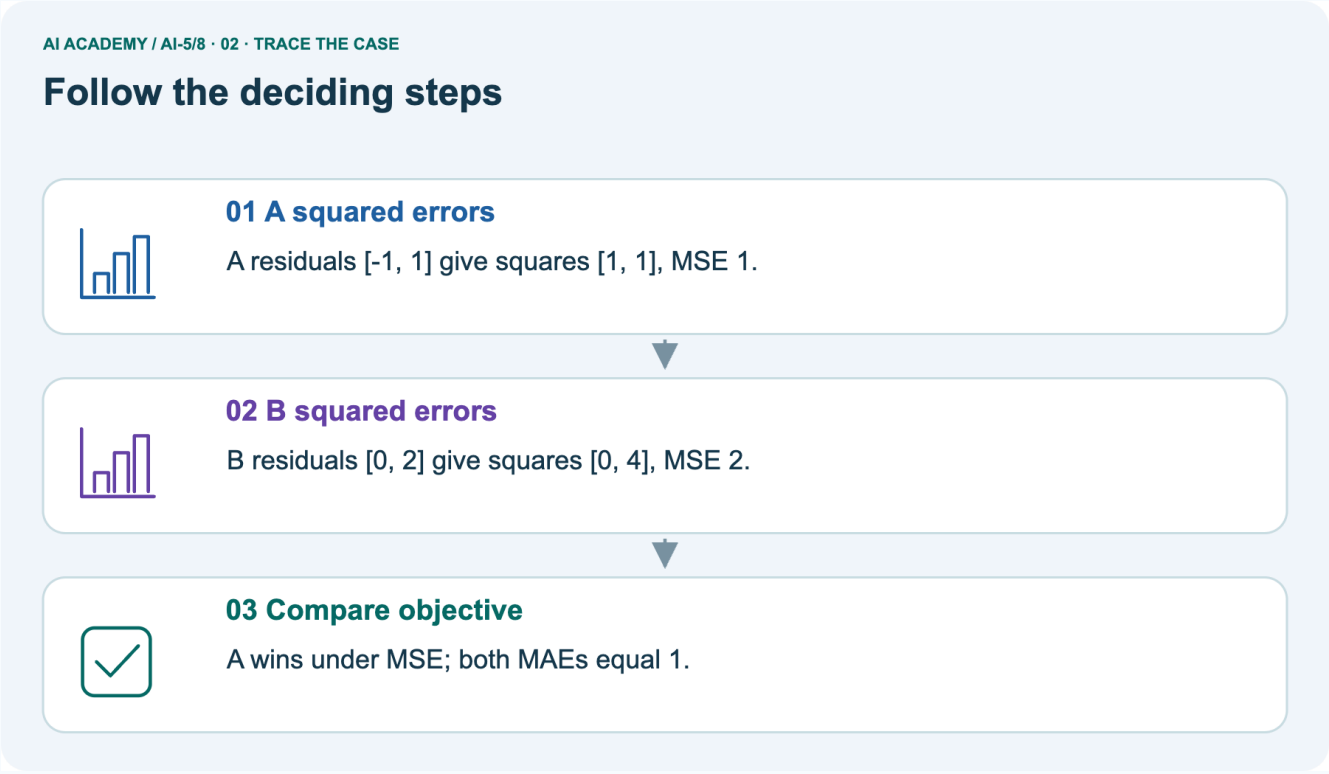
Follow the labelled picture with the matching explanation. Open the vector diagram

Compare candidate losses with the same **target**s and denominator. Supplied facts: **Target**s [1, 3]; A predicts [0, 4]; B predicts [1, 5]. Use MSE over both cases.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

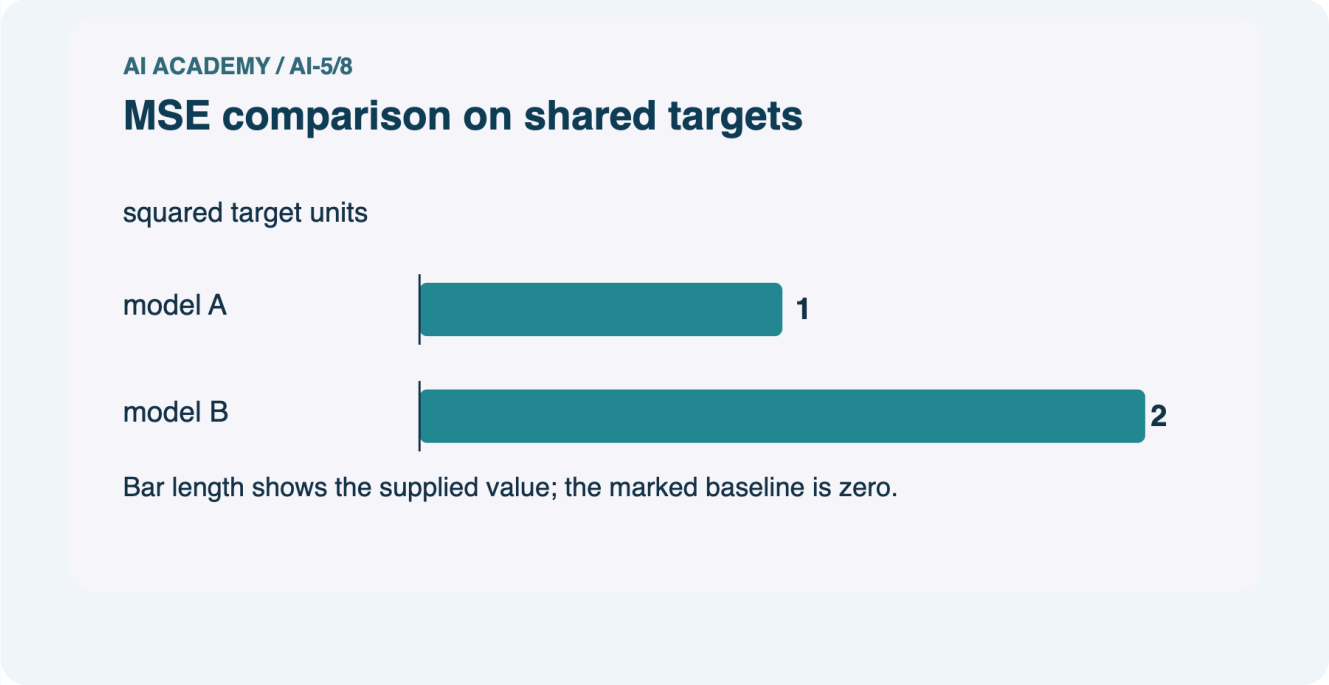


Follow the labelled picture with the matching explanation. Open the vector diagram

A **squared error**s: A **residual**s [-1, 1] give squares [1, 1], MSE 1. → B **squared error**s: B **residual**s [0, 2] give squares [0, 4], MSE 2. → Compare objective: A wins under MSE; both MAEs equal 1.

Point to each numbered stage and explain which recorded **input** it uses.

03. MSE comparison on shared targets

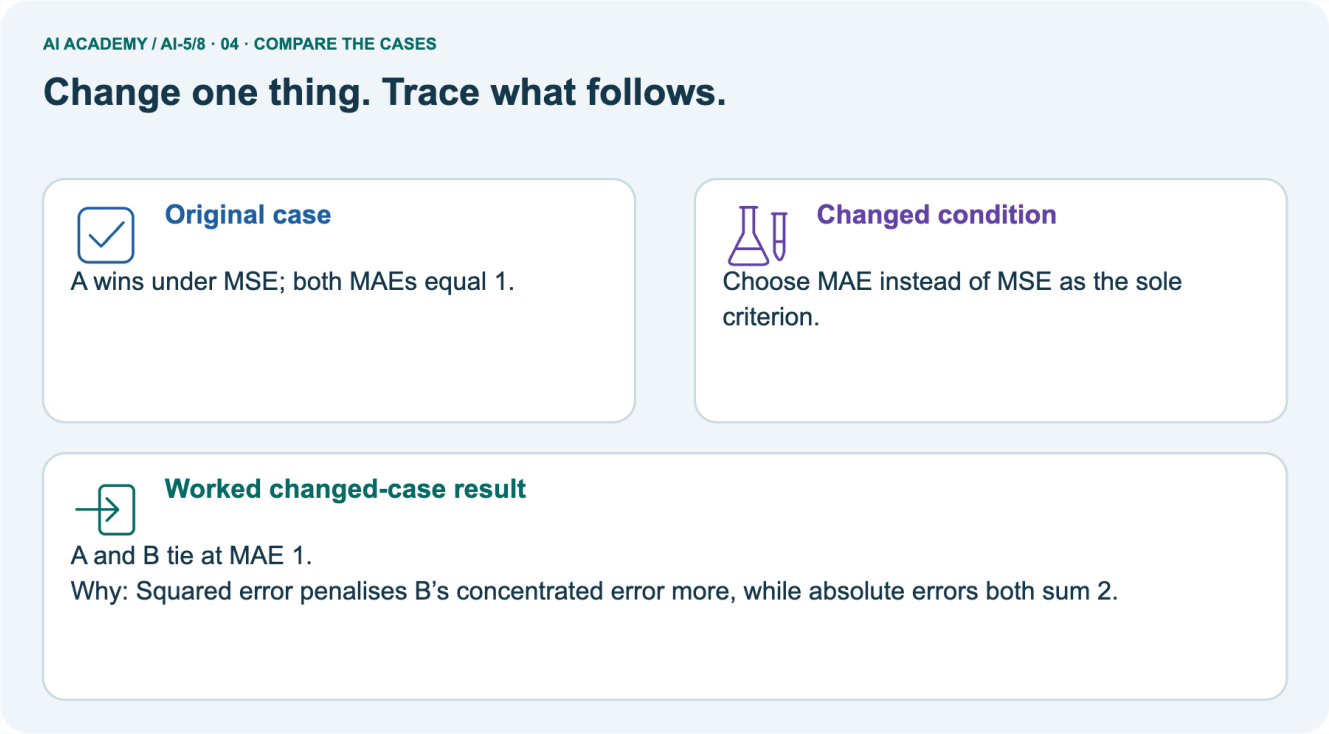


Follow the labelled picture with the matching explanation. Open the vector diagram

MSE comparison on shared **target**s. Units: squared **target** units. model A: 1. model B: 2.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



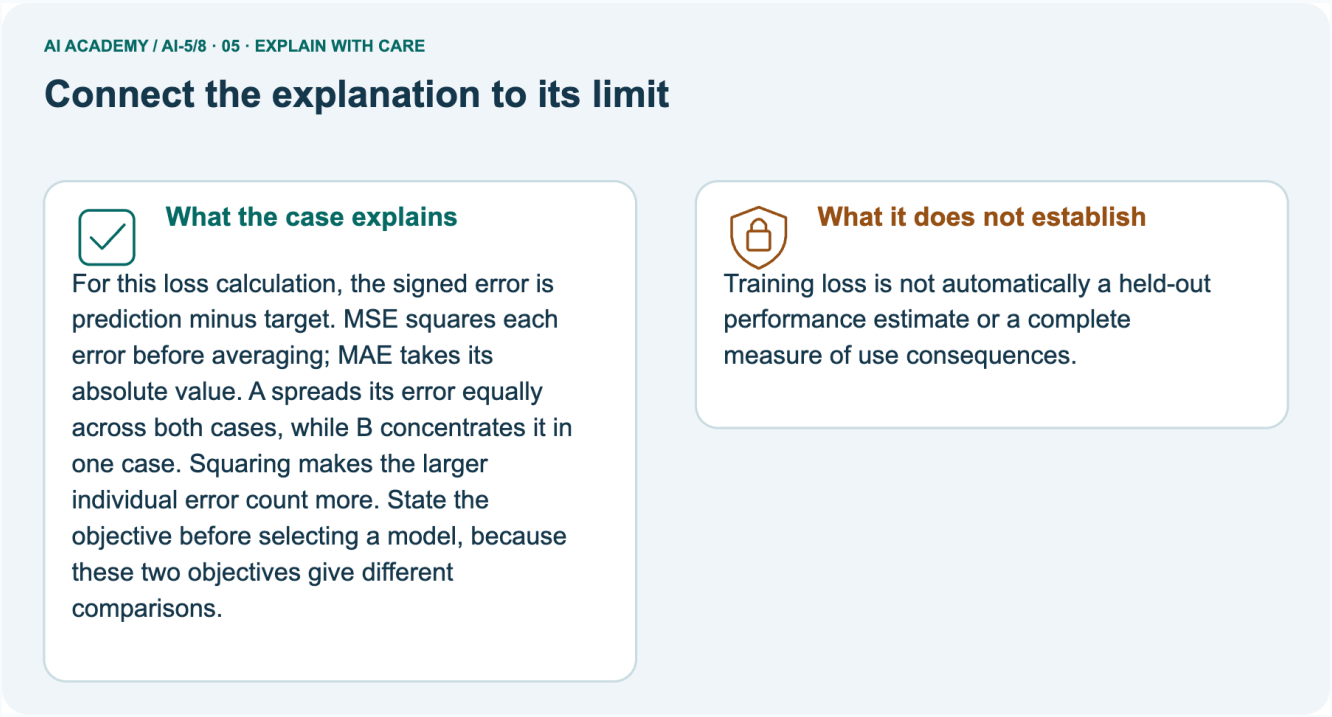
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: A wins under MSE; both MAEs equal 1. Changed condition: Choose MAE instead of MSE as the sole criterion. Expected result: A and B tie at MAE 1.

Reason: **Squared error** penalises B's concentrated error more, while absolute errors both sum 2.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

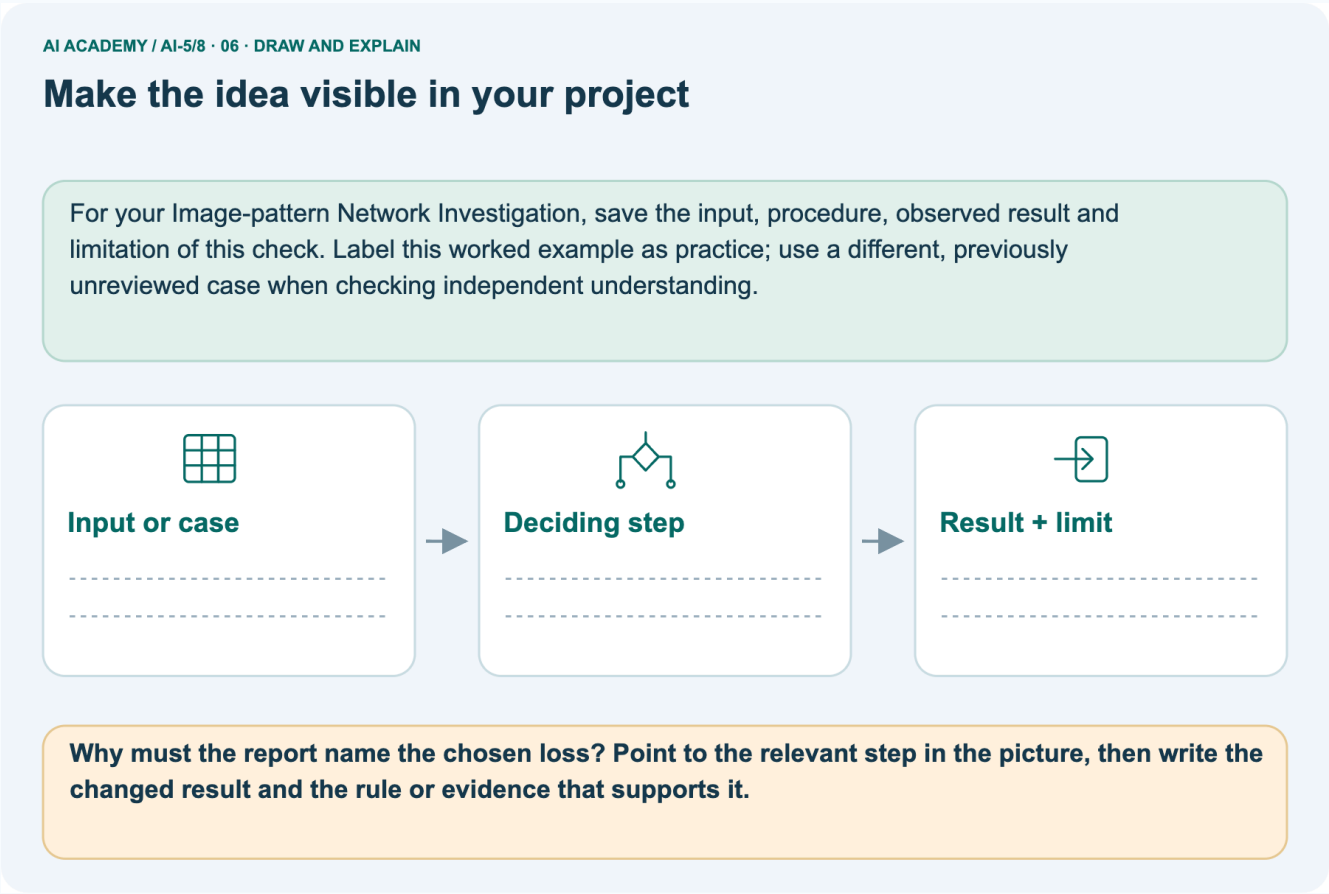
For this loss calculation, the signed error is **prediction** minus **target**. MSE squares each error before averaging; MAE takes its absolute value. A spreads its error equally across both cases, while B concentrates it in one case.

Squaring makes the larger individual error count more. State the objective before selecting a model, because these two objectives give different comparisons.

Limit: **Training** loss is not automatically a held-out performance estimate or a complete measure of use consequences.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why must the report name the chosen loss? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Loss functions: the case

02. Follow the deciding steps

03. MSE comparison on shared targets

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.